

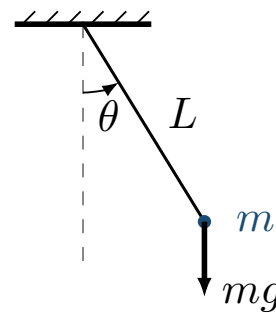
WEEK 1 OVERVIEW

Modeling and qualitative dynamics

Teaching sequence

- 1.1 What makes an oscillator nonlinear?
- 1.2 Physical sources and model derivation
- 1.3 Nondimensionalization
- 1.4 Excitation and energy balance
- 1.5 Equilibria and local linearization
- 1.6 Phase-plane geometry
- 1.7 Energy landscapes and separatrices

One model, several views: forces, equations, local motion, and global geometry.



$$\ddot{\theta} + \frac{g}{L} \sin \theta = 0$$

Exact pendulum model

Dots denote derivatives with respect to t .

1.1 What makes an oscillator nonlinear?

Linearity is a property of the equation

For an operator $\mathcal{L}$, arbitrary x_1, x_2 , and constants a, b :

$$\boxed{\mathcal{L}[ax_1 + bx_2] = a\mathcal{L}[x_1] + b\mathcal{L}[x_2].}$$

Linear oscillator

$$\mathcal{L}[x] = m\ddot{x} + c\dot{x} + kx$$

Cubic oscillator

$$\mathcal{N}[x] = m\ddot{x} + c\dot{x} + kx + \alpha x^3$$

Apply the same test to both operators. What changes when $\alpha \neq 0$?

$$\mathcal{N}[x_1 + x_2] \stackrel{?}{=} \mathcal{N}[x_1] + \mathcal{N}[x_2]$$

$$\neq \alpha(x_1^3) + \alpha(x_2^3)$$

$$m(\ddot{x}_1 + \ddot{x}_2) + c(\dot{x}_1 + \dot{x}_2) + k(x_1 + x_2) + \alpha(x_1 + x_2)^3 = \mathcal{L}[x_1] + \mathcal{L}[x_2] + \alpha(x_1 + x_2)^3$$

1.1 What makes an oscillator nonlinear?

Examples: equation, classification and reason

Equation	Classification	Reason
$\ddot{x} + 2\dot{x} + 3x = \sin t$ <i>→ harmonic excitation</i>	Linear	Forcing depends only on t .
$\ddot{x} + \underbrace{[1 + h \cos(\Omega t)]}_{\text{parametric excitation}} x = 0$	Linear	The coefficient varies with time; x appears to the first power.
$\ddot{x} + x + \alpha x^3 = 0, \alpha \neq 0$	Nonlinear	Cubic dependence on x .
$\ddot{x} + x + \beta \dot{x} \dot{x} = 0, \beta \neq 0$	Nonlinear	Nonlinear dependence on velocity.

Test dependence on the unknown motion, including its derivatives.

Coefficients are time dependent: Parametric excitation.

Illustrative equations with compatible coefficient units; dots denote d/dt .

1.1 What makes an oscillator nonlinear?

Solved example 1: test a cubic oscillator

$$\mathcal{N}[x] = m\ddot{x} + c\dot{x} + kx + \alpha x^3$$

Step 1 Scale the motion

$$\mathcal{N}[2x] = 2m\ddot{x} + 2c\dot{x} + 2kx + \alpha(2x)^3$$

Step 2 Scale the operator output

$$2\mathcal{N}[x] = 2m\ddot{x} + 2c\dot{x} + 2kx + 2\alpha x^3$$

Subtract. When does the difference vanish for every x ?

$$\mathcal{N}[2x] - 2\mathcal{N}[x] = \alpha(2x)^3 - 2\alpha x^3 = 6\alpha x^3 \neq 0$$

Numerical check

$$r(x) = 100x + 10^4 x^3$$

x in metres, r in newtons.

Compare $r(0.10)$ with $2r(0.05)$.

Calculate and interpret.

1.1 What makes an oscillator nonlinear?

Solved example 2: where do higher harmonics come from?

Given

$$r(x) = kx + \alpha x^3, \quad x(t) = A \cos(\Omega t).$$

Step 1 Substitute the prescribed motion

$$r(t) = kA \cos(\Omega t) + \alpha A^3 \cos^3(\Omega t).$$

Step 2 Use the trigonometric identity

$$\cos(3\psi) = 4 \underbrace{\cos^3 \psi} - 3 \cos \psi.$$

$$\begin{aligned} r(t) &= kA \cos(\Omega t) + \alpha A^3 \left(\frac{1}{4} \cos(3\Omega t) + \frac{3}{4} \cos(\Omega t) \right) \\ &= kA \cos(\Omega t) + \frac{3}{4} \alpha A^3 \cos(\Omega t) + \frac{1}{4} \alpha A^3 \cos(3\Omega t) \end{aligned}$$

What to look for

Which frequencies appear in the force?

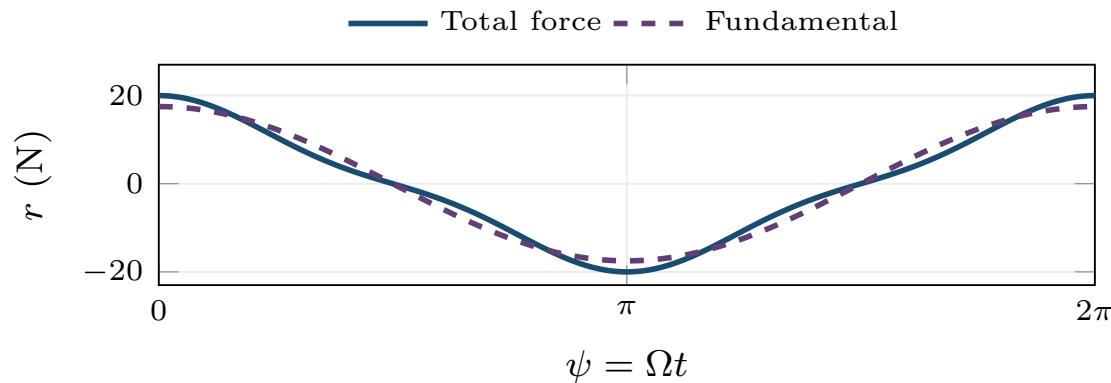
Does a sinusoidal displacement imply a sinusoidal restoring force?

The motion is prescribed here. We are calculating the force.

1.1 What makes an oscillator nonlinear?

The restoring force contains higher harmonics

$$k = 100 \text{ N/m}, \quad \alpha = 10^4 \text{ N/m}^3, \quad A = 0.10 \text{ m.} \quad \leftarrow$$



Calculate

Fundamental force amplitude:

$$kA + \frac{3}{4}\alpha A^3$$

Third-harmonic force amplitude:

$$\frac{1}{4}\alpha A^3$$

Third harmonic / fundamental:

$$14.3\%$$

Interpret this ratio for the force. The displacement amplitude ratio requires a response calculation.

Source connection: Krack–Gross, Sec. 1.2; Thomas, Sec. 4.2.1. Original calculations.

1.2 Physical sources and derivation of nonlinear models

Physical mechanisms and force laws

Source	Mechanism	Example
Geometry	Changing length or direction	$mgL \sin \theta$
Material	Nonlinear elastic response	$\sigma = E\varepsilon + D\varepsilon^3$
Contact	Opening, closing or sliding	$K\langle\delta\rangle_+^{3/2}$
Coupling	Interaction with a surrounding fluid	$F_d = -b \dot{x} \dot{x}$

$\langle\delta\rangle_+ = \max(\delta, 0)$; $K, b > 0$. The elastic law is illustrative.

Which examples store energy? Which dissipate it?

$$f(t) = K\langle\delta\rangle_+^{3/2} \rightarrow V(t) = \frac{2}{5}K\langle\delta\rangle_+^{5/2}$$

Thomas, Secs. 4.2.2–4.2.3.

1.2 Physical sources and derivation of nonlinear models

Energy storage and dissipation: worked classification

Storage: work can be recovered in these idealized models

Pendulum $V = mgL(1 - \cos \theta)$, so $dV/d\theta = mgL \sin \theta$.

Elastic material $W(\varepsilon) = \frac{1}{2}E\varepsilon^2 + \frac{1}{4}D\varepsilon^4$, so $dW/d\varepsilon = \sigma$. W is energy per unit volume.

Elastic contact $V_c(\delta) = \frac{2}{5}K\langle\delta\rangle_+^{5/2}$, so $dV_c/d\delta = K\langle\delta\rangle_+^{3/2}$.

Loss: the force does negative work on the motion

$$F_d \dot{x} = -b|\dot{x}|^3 \leq 0, \quad F_{\text{fr}} \dot{x} = -\mu N |\dot{x}| \leq 0.$$

The listed elastic contact law stores energy. Frictional contact or an inelastic impact can dissipate it. The physical category alone does not determine energy behavior.

1.2 Physical sources and derivation of nonlinear models

A modeling sequence and its assumptions

Assumptions Coordinates Geometry Forces Balance

A common structural-vibration model is

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} + \mathbf{f}_{\text{nl}}(\mathbf{q}, \dot{\mathbf{q}}) = \mathbf{f}(t).$$

1 DoF

$$m\ddot{x} + c\dot{x} + kx + \alpha x^3 = f(t)$$

Sliding friction: check the power

$$F_{\text{fr}} = -\mu N \operatorname{sgn}(\dot{x}), \quad \dot{x} \neq 0.$$

Calculate $F_{\text{fr}}\dot{x}$. What must change at rest?

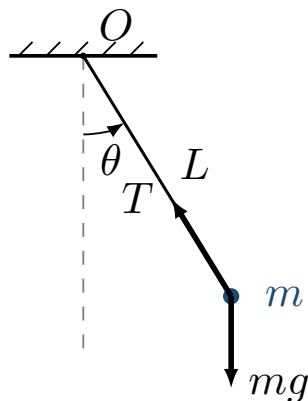
$$F_{\text{fr}}\dot{x} = -\mu N |\dot{x}| ;$$

At rest, $F_{\text{fr}} \in [-\mu N, \mu N]$ can enforce sticking (equal static/kinetic idealization).

Thomas, Sec. 4.2.3; Touzé–Vizzaccaro, Sec. 2, Eqs. (1)–(2).

1.2 Physical sources and derivation of nonlinear models

Solved example 3: derive the exact pendulum equation



Given

Point mass, massless rigid rod, fixed frictionless pivot, uniform gravity. No damping or applied torque.

1. Write the inertia about O .
2. Write the gravity moment.
3. Apply the moment balance.

$$I_O = mL^2, \quad \sum M_O = -mgL \sin \theta$$

$$\leftarrow -mgL \sin \theta = I_O \ddot{\theta} = mL^2 \ddot{\theta}$$

The rod force passes through the pivot. Use radians and measure θ from the downward vertical.

$$\ddot{\theta} + \frac{g}{L} \sin \theta = 0$$

eq. of motion!

1.2 Physical sources and derivation of nonlinear models

Pendulum: a local approximation and its error

$$\ddot{\theta} + \frac{g}{L} \sin \theta = 0, \quad \sin \theta = \theta - \frac{\theta^3}{6} + O(\theta^5).$$

→ Taylor series.

Linear and cubic models

Check at $30^\circ = \pi/6$

Retain the appropriate terms.

State the small-angle assumption.

$$e_{\text{lin}} = \frac{|\theta - \sin \theta|}{|\sin \theta|}.$$

linear : $\ddot{\theta} + \frac{g}{L} \theta = 0$
cubic : $\ddot{\theta} + \frac{g}{L} \theta - \frac{g}{6L} \theta^3 = 0$
exact : $\ddot{\theta} + \frac{g}{L} \sin \theta = 0$

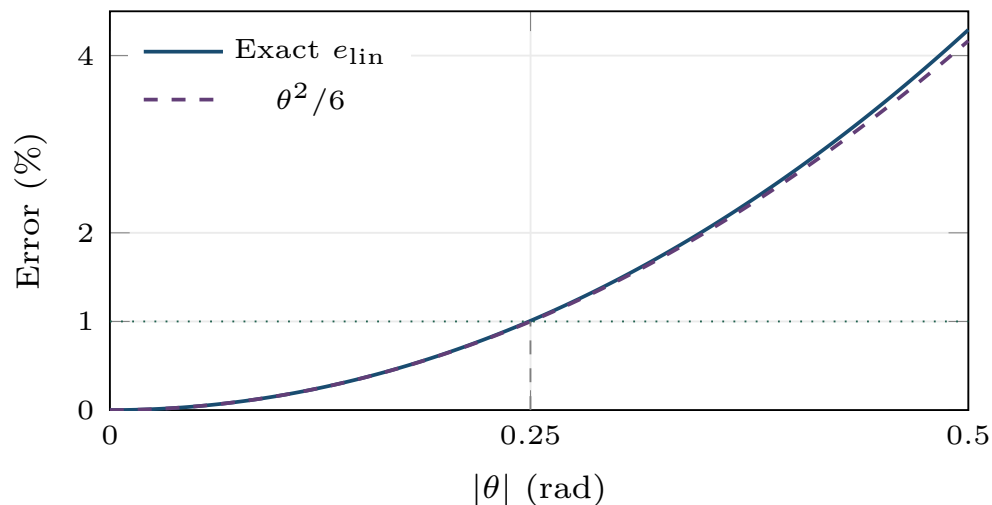
Calculate the error, then repeat with the cubic approximation.

Use the exact torque in the denominator.

Torque error and period error measure different things. A tolerance determines the acceptable angle.

1.2 Physical sources and derivation of nonlinear models

Pendulum: a 1% torque-error tolerance



Near zero

$$e_{\text{lin}} = \left| \frac{\theta}{\sin \theta} - 1 \right| \sim \frac{\theta^2}{6}.$$

Leading estimate

$$\frac{\theta^2}{6} \leq 0.01$$

$$|\theta| \lesssim \sqrt{0.06}$$

$$|\theta| \lesssim 0.245 \text{ rad} \approx 14^\circ.$$

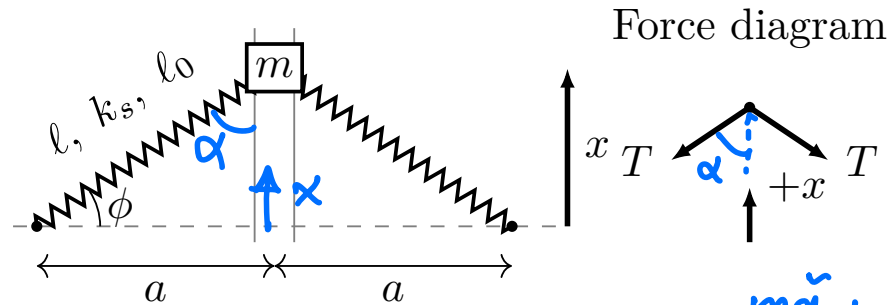
This is an asymptotic estimate. The exact threshold is about 0.2441 rad.

At 30°: linear torque error 4.7198%; cubic error 0.065164%.

1.2 Physical sources and derivation of nonlinear models

Solved example 4: linear springs, nonlinear oscillator

Two identical axial springs: stiffness k_s , unstretched length $\ell_0 \leq a$. The mass slides along a straight guide in a horizontal plane.



$$-2T \sin \phi = m \ddot{x}$$

$$m \ddot{x} + 2T \sin \phi = 0$$

$$m \ddot{x} + 2K_s \frac{(\sqrt{a^2 + x^2} - \ell_0)x}{\sqrt{a^2 + x^2}} = 0$$

Find the spring length and tension, then project both forces along x .

$$\ell(x) = \sqrt{a^2 + x^2}, \quad T(x) = K_s (\ell(x) - \ell_0), \quad \sin \phi = \frac{x}{\ell(x)}$$

1.2 Physical sources and derivation of nonlinear models

Spring example: exact equation and energy check

$$\ell(x) = \sqrt{a^2 + x^2}, \quad T = k_s(\ell - \ell_0), \quad F_x = -2T \frac{x}{\ell}.$$

Step 1 Apply Newton's law

$m\ddot{x} = F_x$ Write the restoring term $r(x)$ on the left.

$$m\ddot{x} + 2k_s \left(1 - \frac{\ell_0}{\sqrt{a^2 + x^2}}\right)x = 0$$

$$\underbrace{m\ddot{x}}_{\text{inertia part}} + \underbrace{2k_s x - \frac{\ell_0}{\sqrt{a^2 + x^2}}x}_{\text{restoring part (stiffness)}} = 0$$

For $x > 0$, both spring components oppose displacement. Check that $r(-x) = -r(x)$ and $r(0) = 0$.

$$r(x) = 2k_s \left(1 - \frac{\ell_0}{\sqrt{a^2 + x^2}}\right)x$$

1.2 Physical sources and derivation of nonlinear models

Spring example: an independent energy check

$$\ell(x) = \sqrt{a^2 + x^2}, \quad r(x) = 2k_s \left(1 - \frac{\ell_0}{\ell}\right) x.$$

Differentiate the stored energy

$$V(x) = 2 \left[\frac{1}{2} k_s (\ell - \ell_0)^2 \right].$$

$\int dx$ $\frac{dV}{dx}$

Differentiate V with respect to x and compare with $r(x)$.

$$\frac{dV}{dx} = 2 \left[k_s (\ell - \ell_0) \cdot \frac{d\ell}{dx} \right] = 2k_s \left(1 - \frac{\ell_0}{\ell}\right) \frac{\ell \cdot 2x}{2\ell} = 2k_s \left(1 - \frac{\ell_0}{\ell}\right) x$$

Check signs, units and the factor two. Geometry makes the restoring term nonlinear.

1.2 Physical sources and derivation of nonlinear models

Spring example: cubic approximation and numerical check

$$2k_s \left(1 - \frac{\ell_0}{a} \left(1 - \frac{1}{2} \left(\frac{x^2}{a^2}\right)\right)\right) x$$
$$\underbrace{\left(2k_s - \frac{\ell_0}{a}\right)}_{k_{\text{eff}}} x$$
$$r(x) = 2k_s \left(1 - \frac{\ell_0}{\sqrt{a^2 + x^2}}\right) x, \quad |x|/a \ll 1.$$
$$\frac{1}{\sqrt{a^2 + x^2}} = \frac{1}{a} \left[1 - \frac{1}{2} (x/a)^2 + O((x/a)^4)\right].$$

Identify the coefficients

$$r(x) = k_{\text{eff}} x + \alpha_{\text{eff}} x^3 + O(x^5).$$

$$k_{\text{eff}} = 2k_s - \frac{\ell_0}{a} \quad \alpha_{\text{eff}} = \frac{k_s \ell_0}{a^3}$$

What changes when $\ell_0 = a$?

Models: Nayfeh–Mook, Sec. 2.1.1; Thomas, Sec. 4.2.2. Original spring derivation.

$$+ \frac{2k_s \ell_0}{2a} \frac{x^3}{a^2}$$
$$\left(k_s \ell_0 \frac{x^3}{a^3} \right) \alpha_{\text{eff}}$$

Numerical check

$m = 1$ kg, $a = 0.50$ m, $\ell_0 = 0.40$ m, $k_s = 100$ N/m.

At $x = 0.10$ m, compare the exact and cubic restoring forces.

Calculate the relative force error.

1.3 Nondimensionalization and measures of nonlinearity

Displacement and time scales

For $X, T > 0$, choose $x(t) = Xu(\tau)$ and $\tau = t/T$.

Step 1 Apply the chain rule

$$\dot{x} = X \frac{du}{d\tau} \frac{d\tau}{dt},$$

$$\dot{x} = \frac{X}{T} u', \quad \ddot{x} = \frac{X}{T^2} u''.$$

Step 2 Transform the initial data

$$x(0) = x_0 \rightarrow u(0) = \frac{x_0}{X} \quad u'(0) = \frac{T v_0}{X}$$

Check: $u(0) = x_0/X$, $u'(0) = Tv_0/X$.

Dots mean d/dt ; primes on time histories mean $d/d\tau$. X is a reference displacement, not necessarily the response amplitude. Scaling does not discard terms.

1.3 Nondimensionalization and measures of nonlinearity

Scaling the forced cubic oscillator

$$m\ddot{x} + c\dot{x} + kx + \alpha x^3 = F \cos(\Omega t), \quad m, k > 0.$$

Substitute $x = Xu$, $t = T\tau$ and divide by kX

$$\frac{m}{kT^2}u'' + \frac{c}{kT}u' + u + \frac{\alpha X^2}{k}u^3 = \frac{F}{kX} \cos(\Omega T\tau).$$

Choose $T = \sqrt{m/k} = 1/\omega_0$, where $\omega_0 = \sqrt{k/m}$.

$$u'' + 2\zeta u' + u + \varepsilon u^3 = p \cos(\rho\tau)$$

$$\zeta = \frac{c}{2\sqrt{mk}}, \quad \varepsilon = \frac{\alpha X^2}{k}, \quad p = \frac{F}{kX}, \quad \rho = \frac{\Omega}{\omega_0}.$$

All four coefficients are dimensionless. The symbol ε does not by itself mean the coefficient is small.

$$m \frac{X}{T^2} u'' + c \frac{X}{T} u' + kXu + \alpha X^3 u^3 = F \cos(\Omega T\tau)$$

$\downarrow \div kX$
 $\frac{c}{kT} \cdot \frac{kT^2}{m} = \frac{cT}{m}$
 $\frac{c}{\sqrt{mk}}$

1.3 Nondimensionalization and measures of nonlinearity

Solved example 5: the geometric spring oscillator

$$m = 1 \text{ kg}, \quad k = 40 \text{ N/m}, \quad \alpha = 320 \text{ N/m}^3, \quad X = 0.10 \text{ m},$$

$$c = 0.40 \text{ N s/m}, \quad F = 2 \text{ N}, \quad \Omega = 5 \text{ rad/s}.$$

Calculate ω_0 , 2ζ , ε , p , and ρ .

Result check

$$\omega_0 = 6.32456 \text{ rad/s}, \quad T = 0.158114 \text{ s},$$

$$u'' + 0.0632456u' + u + 0.0800u^3 = 0.5000 \cos(0.790569\tau).$$

For $x(0) = 0.050 \text{ m}$, $\dot{x}(0) = 0$: $u(0) = 0.5$, $u'(0) = 0$. At $\tau = 2$, $t = 0.316228 \text{ s}$.

1.3 Nondimensionalization and measures of nonlinearity

Nonlinearity depends on the motion amplitude

$$\eta(x) = \frac{|\alpha x^3|}{|kx|} = \frac{|\alpha| x^2}{k} = |\varepsilon| u^2 \quad (x \neq 0); \quad \eta(0) = 0.$$

$ x $ (m)	$ u $	η	Cubic / linear force
0.05	0.5	0.02	2%
0.10	1	0.08	8%
0.30	3	0.72	72%

$$\varepsilon = \frac{\alpha X^2}{K}$$
$$\tilde{\varepsilon} = \frac{\alpha s^2 X^2}{K}$$
$$\tilde{\varepsilon} = \frac{\alpha \tilde{X}^2}{K}$$

Does choosing a larger reference scale X make the physical motion more nonlinear?

Set $\tilde{X} = sX$ and compare $|\tilde{\varepsilon}| \tilde{u}^2$ with $|\varepsilon| u^2$. $\tilde{X} = sX$

No: $\tilde{u} = u/s$, $\tilde{\varepsilon} = s^2 \varepsilon$. The force ratio is unchanged. Weak nonlinearity requires $|\varepsilon| \max |u|^2 \ll 1$.

1.3 Nondimensionalization and measures of nonlinearity

Solved example 6: the scaled pendulum

$$\ddot{\theta} + \frac{g}{L} \sin \theta = 0, \quad \theta = \Theta u, \quad \tau = \sqrt{g/L} t, \quad \Theta > 0.$$

Substitute and divide by $\Theta g/L$. Transform a release from $\theta(0) = \Theta$.

$$u'' + \frac{\sin(\Theta u)}{\Theta} = 0, \quad u(0) = 1, \quad u'(0) = 0.$$

For $|\Theta u| \ll 1$:

$$u'' + u - \frac{\Theta^2}{6} u^3 \approx 0, \quad \varepsilon = -\Theta^2/6.$$

At $\Theta = 30^\circ = \pi/6$, $\varepsilon = -0.0456926$. Scaling is exact; truncating the sine series introduces the approximation.

1.3 Nondimensionalization and measures of nonlinearity

Extension: when linear stiffness vanishes

For the spring mechanism with $\ell_0 = a$, the leading local model is

$$m\ddot{x} + \alpha x^3 = 0, \quad \alpha = k_s/a^2 > 0.$$

Balance inertia against the surviving restoring term

$$\frac{mX}{T^2} = \alpha X^3.$$

$$\frac{mX}{T^2} u'' + \alpha X^3 u^3 = 0$$

↓

$$\underbrace{u'' + u^3}_{=0} = 0$$

Find T and substitute $x = Xu$, $\tau = t/T$.

$$\dot{x} = \frac{X}{T} u'$$

$$\ddot{x} = \frac{X}{T^2} u''$$

$$T = \sqrt{\frac{m}{\alpha X^2}}, \quad u'' + u^3 = 0.$$

The frequency scale $1/T = X\sqrt{\alpha/m}$ depends on amplitude. It is not the exact oscillation frequency. The cubic mechanism approximation still requires $|x|/a \ll 1$.

1.4 Autonomous, forced, conservative and dissipative systems

Time dependence and excitation mechanism

$$\dot{\mathbf{z}} = \mathbf{f}(\mathbf{z}) \quad \text{autonomous,} \quad \dot{\mathbf{z}} = \mathbf{f}(\mathbf{z}, t) \quad \text{nonautonomous.}$$

Mechanism	Example
Direct forcing	$m\ddot{x} + c\dot{x} + kx + \alpha x^3 = F \cos(\Omega t)$
Parametric excitation	$m\ddot{x} + c\dot{x} + k(t)x = 0$
Self-excitation	$u'' - \mu(1 - u^2)u' + u = 0, \mu > 0, \text{ primes mean } d/d\tau$

Can an autonomous system lose or gain mechanical energy?

$$m\ddot{x} + c\dot{x} + kx + \alpha x^3 = f \quad \mathbf{z} = \begin{Bmatrix} x \\ \dot{x} \end{Bmatrix} \quad \dot{\mathbf{z}} = \begin{Bmatrix} \dot{x} \\ \ddot{x} \end{Bmatrix} = \begin{Bmatrix} z(2) \\ \frac{f}{m} - \frac{c}{m}z(2) - \frac{k}{m}z(1) - \frac{\alpha}{m}z(1)^3 \end{Bmatrix} = \mathbf{f}(\vec{z})$$

Yes. Free viscous damping removes energy. The van der Pol model supplies energy in part of state space. Autonomy describes explicit time dependence.

1.4 Autonomous, forced, conservative and dissipative systems

The mechanical energy balance

For constant $m > 0$ and time-independent $V(x)$,

$$m\ddot{x} + \frac{dV}{dx} + d(x, \dot{x}) = f(t).$$

Multiply by velocity and use the chain rule

$$m\dot{x}\ddot{x} + \frac{dV}{dx}\dot{x} = f(t)\dot{x} - d(x, \dot{x})\dot{x}.$$

Identify the derivative of kinetic energy and the derivative of potential energy.

$$E = \frac{1}{2}m\dot{x}^2 + V(x), \quad \dot{E} = f\dot{x} - d\dot{x}.$$

$f\dot{x}$ is input power. $d\dot{x} \geq 0$ is passive loss. If $f = d = 0$, energy is conserved. A negative value of $d\dot{x}$ represents energy supply.

1.4 Autonomous, forced, conservative and dissipative systems

Solved example 7: classify the Duffing oscillator

$\int kx\dot{x} + \alpha x^3\dot{x} dt \rightarrow V = \frac{1}{2}kx^2 + \frac{1}{4}\alpha x^4, \quad \dot{E} = F \cos(\Omega t)\dot{x} - c\dot{x}^2.$

$\star m\ddot{x} + c\dot{x} + kx + \alpha x^3 = F \cos(\Omega t), \quad m, k > 0, \quad c \geq 0, \quad \Omega > 0.$

$\int F \cos(\Omega t) \dot{x} - c\dot{x}^2 dt$

Parameters	Time dependence	Energy
$c = F = 0$	<u>Autonomous</u>	<u>Conserved</u>
$c > 0, F = 0$	<u>Autonomous</u>	Nonincreasing: <u>$\dot{E} = -c\dot{x}^2$</u> loss of energy
$c = 0, F \neq 0$	<u>Nonautonomous</u>	May increase or decrease depend on $f(t)$
$c > 0, F \neq 0$	Nonautonomous	Input competes with damping loss $f(t), c \uparrow \downarrow$

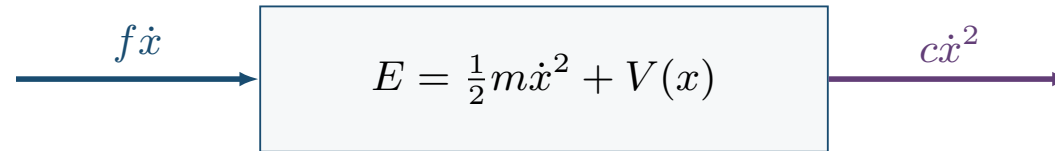
At a turning point, $\dot{E} = 0$. Does this prove conservation?

No. Conservation requires E to remain constant along the motion, not just have zero rate at one instant.

1.4 Autonomous, forced, conservative and dissipative systems

Duffing example: calculate instantaneous power

$$c = 0.40 \text{ N s/m}, \quad F = 2 \text{ N}, \quad \cos(\Omega t) = 1, \quad \dot{x} = 0.30 \text{ m/s}.$$



*Nonautonomous
with $\dot{E} > 0$*

$$P_{\text{in}} = f\dot{x} = 2(0.3) \quad P_{\text{loss}} = c\dot{x}^2 = 0.4(0.3)^2 \quad \dot{E} = P_{\text{in}} - P_{\text{loss}} = 0.564 \text{ W} > 0$$

$$P_{\text{in}} = 0.600 \text{ W}, \quad P_{\text{loss}} = 0.036 \text{ W}, \quad \boxed{\dot{E} = 0.564 \text{ W}.}$$

With velocity reversed at the same forcing phase, $\dot{E} = -0.636 \text{ W}$. These are evaluations at specified states, not a computed trajectory.

1.4 Autonomous, forced, conservative and dissipative systems

Solved example 8: time-dependent stiffness

$$m\ddot{x} + k(t)x = 0, \quad k(t) = k_0[1 + h \cos(\Omega t)],$$

$$k_0 > 0, \quad 0 < |h| < 1, \quad \Omega > 0.$$

Differentiate the time-dependent energy

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}k(t)x^2.$$

Apply the product rule, including the derivative of $k(t)$.

linear, nonautonomous, parametric excitation!

$$\dot{E} = \frac{1}{2}\dot{k}(t)x^2 = -\frac{1}{2}k_0h\Omega \sin(\Omega t)x^2.$$

Linear, nonautonomous and parametrically excited. For $k_0 = 40$ N/m, $h = 0.2$, $\Omega = 5$ rad/s, $x = 0.1$ m and $\sin \Omega t = -1$: $\dot{E} = 0.20$ W. Positive instantaneous power alone does not prove instability.

1.4 Autonomous, forced, conservative and dissipative systems

Solved example 9: self-excitation and energy feedback

$$u'' - \mu(1 - u^2)u' + u = 0, \quad \mu > 0, \quad u' = du/d\tau.$$

$$\mathcal{E} = \frac{1}{2}(u')^2 + \frac{1}{2}u^2, \quad \mathcal{E}' = u'(u'' + u).$$

Substitute the equation of motion and determine the sign of $\mathcal{E}'$.

$$\mathcal{E}' = \mu(1 - u^2)(u')^2.$$

$ u < 1, u' \neq 0$	$\mathcal{E}' > 0$	Energy supplied
$ u > 1, u' \neq 0$	$\mathcal{E}' < 0$	Energy removed
$ u = 1$ or $u' = 0$	$\mathcal{E}' = 0$	Instantaneous zero rate

For $\mu = 0.2$: $(u, u') = (0.5, 1)$ gives $\mathcal{E}' = 0.15$; $(2, 1)$ gives -0.60 . Rates are dimensionless. The sign pattern alone does not prove a stable limit cycle. A physical source supplies the energy.

1.5 Equilibria and local linearization

Force balance: a motion that remains at rest

$$m\ddot{x} + c\dot{x} + r(x) = F_0, \quad m > 0, \quad c \geq 0.$$

Here F_0 is constant and $r(x)$ is smooth.

Step 1 Require rest for all time

equilibrium $\leftarrow x(t) = x_e \implies \dot{x} = \ddot{x} = 0.$

$$m(0) + c(0) + r(x_e) = F_0 \rightarrow r(x_e) = F_0$$

$$r(x_e) = F_0.$$

Is zero velocity at one instant sufficient?

No. If $r(x) \neq F_0$, acceleration is nonzero: the mass cannot remain at rest. A turning point need not be an equilibrium.

1.5 Equilibria and local linearization

Shift first, then linearize

$$mg = k \underline{x_e^3}$$

Step 2 Measure a small displacement from each equilibrium

$$kx^3 = k(\xi + x_e)^3$$

$$x = x_e + \xi, \quad \dot{x} = \dot{\xi}, \quad \ddot{x} = \ddot{\xi}.$$

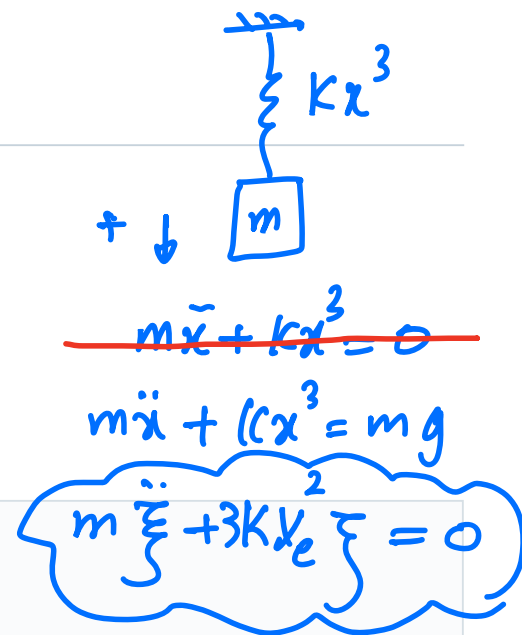
$$r(x_e + \xi) = r(x_e) + k_t \xi + O(\xi^2), \quad k_t = \left. \frac{dr}{dx} \right|_{x_e}.$$

$$Kx^3 \rightarrow K_{eq}\xi; \quad K_{eq} = \left. \frac{df}{dx} \right|_{x=x_e} = 3Kx_e^2$$

$$m\ddot{\xi} + c\dot{\xi} + k_t\xi = 0.$$

k_t is the tangent stiffness at the chosen equilibrium, not the secant $r(x_e)/x_e$. Repeat the calculation at every equilibrium.

The force-balance equation is exact. The perturbation equation is a local approximation. Dots still mean derivatives with respect to physical time t .



1.5 Equilibria and local linearization

Growth rates and local stability

Try $\xi = Ce^{\lambda t}$ in the linear perturbation equation:

$$m\lambda^2 + c\lambda + k_t = 0,$$

$$\lambda_{\pm} = \frac{-c \pm \sqrt{c^2 - 4mk_t}}{2m}.$$

Case	Conclusion for this smooth mechanical model
$k_t > 0, c > 0$	Negative real parts: locally asymptotically stable.
$k_t < 0, c \geq 0$	One positive root: unstable.
$k_t > 0, c = 0$	$\lambda = \pm i\sqrt{k_t/m}$; stable by the energy minimum, without attraction.
$k_t = 0$	A zero root: linearization alone is inconclusive.

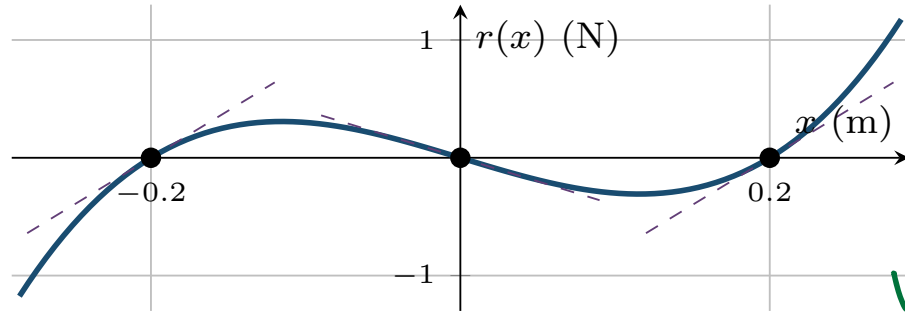
For $c = 0$, use $U(x) = \int r(x) dx - F_0 x$. At equilibrium, $dU/dx = 0$ and $d^2U/dx^2 = k_t$. A strict local minimum gives stability.

1.5 Equilibria and local linearization

Solved example 10: three equilibria, three tangents

$$m\ddot{x} + c\dot{x} - ax + bx^3 = 0, \quad m = 1 \text{ kg}, \quad c = 0.40 \text{ N s/m}, \quad a = 4 \text{ N/m}, \quad b = 100 \text{ N/m}^3.$$

$\int \leftarrow x_e(-a + bx_e^2) = 0 \implies \boxed{x_e = 0, \pm 0.20 \text{ m.}}$



Evaluate each slope

$$k_t = -a + 3bx_e^2.$$

$$k_t(0) = -4 \text{ N/m},$$

$$k_t(\pm 0.20) = 8 \text{ N/m}.$$

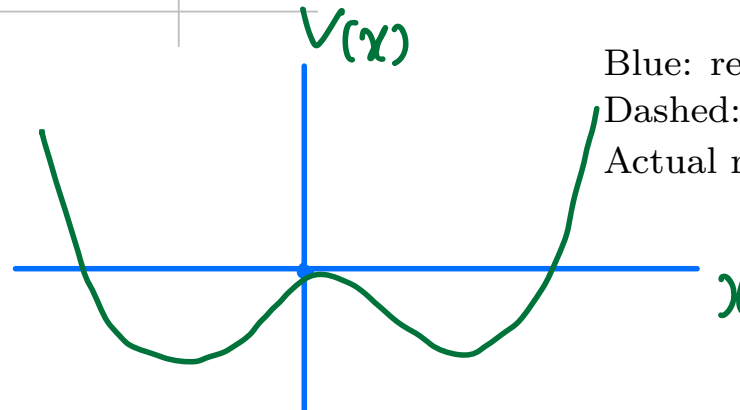
Blue: restoring term $r(x)$.

Dashed: local tangents.

Actual restoring force: $-r(x)$.

$$\int -ax \dot{x} + bx^3 \dot{x} dt$$

$$V(x) = -\frac{ax^2}{2} + \frac{1}{4}bx^4$$



1.5 Equilibria and local linearization

Cubic example: calculate the local motion

At the origin: $k_t = -4 \text{ N/m}$

$$\ddot{\xi} + 0.40\dot{\xi} - 4\xi = 0, \quad \lambda_{\pm} = 1.80998, -2.20998 \text{ s}^{-1}.$$

One mode grows: $x_e = 0$ is unstable.

At either outer equilibrium: $k_t = 8 \text{ N/m}$

Solve $\lambda^2 + 0.40\lambda + 8 = 0$. Write the decaying oscillation.

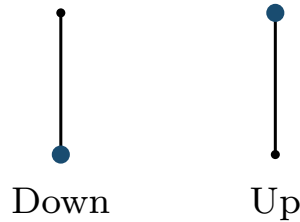
$$\lambda_{\pm} = (-0.20 \pm 2.82135 i) \text{ s}^{-1},$$

$$x(t) \approx x_e + e^{-0.20t} [C_1 \cos(2.82135t) + C_2 \sin(2.82135t)].$$

Both outer equilibria are locally asymptotically stable. t is in seconds; C_1, C_2 follow from small perturbation initial data. Local analysis does not predict the destination of a large disturbance.

1.5 Equilibria and local linearization

Solved example 11: downward and upward pendulum



$$\ddot{\theta} + \frac{g}{L} \sin \theta = 0, \quad \theta_e = n\pi.$$

Set $\theta = \theta_e + \eta$, with $|\eta| \ll 1$ radians:

$$\sin(\theta_e + \eta) = \cos \theta_e \eta + O(\eta^3).$$

$$\ddot{\eta} + \frac{g}{L} \cos \theta_e \eta = 0.$$

Write the local equation at $\theta_e = 0$ and at $\theta_e = \pi$.

Down: $\ddot{\eta} + (g/L)\eta = 0$, stable by the energy minimum, no attraction.

Up: $\ddot{\eta} - (g/L)\eta = 0$, an exponentially growing mode, unstable.

1.5 Equilibria and local linearization

Pendulum example: physical and scaled time

For $L = 1.00$ m and $g = 9.81$ m/s²:

$$\omega_0 = \sqrt{g/L} = 3.13209 \text{ s}^{-1}, \quad \lambda_{\text{down}} = \pm i\omega_0, \quad \lambda_{\text{up}} = \pm \omega_0.$$

Change the time variable explicitly

$$\tau = \omega_0 t, \quad \eta(t) = q(\tau).$$

Apply the chain rule twice, then scale both local equations.

$$\dot{\eta} = \omega_0 q', \quad \ddot{\eta} = \omega_0^2 q'',$$

$$q'' + q = 0 \text{ (down)}, \quad q'' - q = 0 \text{ (up)}.$$

Dots mean d/dt; primes mean d/d τ . The dimensionless growth rates are $\pm i$ and ± 1 , respectively.

1.5 Equilibria and local linearization

Zero tangent stiffness: retain the nonlinear terms

Compare two undamped models with $m, \alpha > 0$:

$$m\ddot{x} + \alpha x^3 = 0 \quad \text{and} \quad m\ddot{x} - \alpha x^3 = 0.$$

Both have $x_e = 0$, $k_t = 0$, and the same linearization $\ddot{\xi} = 0$.

Use energy for the plus sign; release from $x(0) > 0$, $\dot{x}(0) = 0$ for the minus sign.

Plus sign: a strict energy minimum

$$E = \frac{1}{2}m\dot{x}^2 + \frac{1}{4}\alpha x^4 \implies \text{stable, without attraction.}$$

Minus sign: acceleration away from equilibrium

$$\ddot{x} = \frac{\alpha}{m}x^3 > 0 \quad (x > 0) \implies \text{unstable.}$$

Zero or imaginary growth rates do not settle nonlinear stability in general. Use additional structure, such as energy. Next: phase-plane geometry in 1.6.

1.5 Equilibria and local linearization

Name the equilibrium type

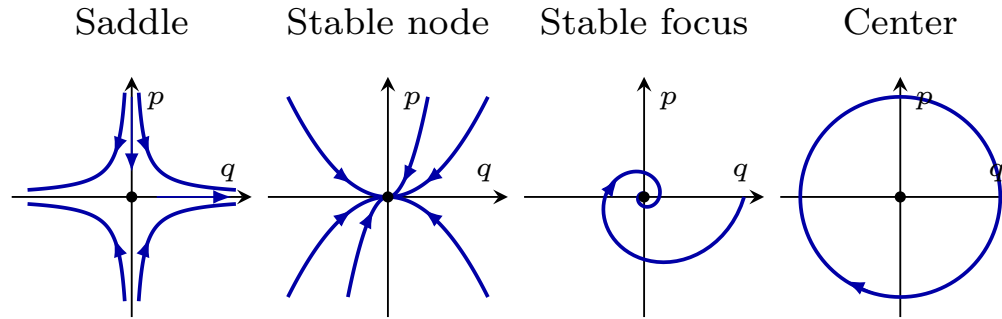
For a smooth planar system, shift to an equilibrium: $\mathbf{y} = \mathbf{z} - \mathbf{z}_e$.
$$\dot{\mathbf{y}} = A\mathbf{y} + O(\|\mathbf{y}\|^2), \quad A = D\mathbf{f}(\mathbf{z}_e).$$

Type	Eigenvalues	Behavior
Saddle	Real, opposite signs	One attracting direction and one repelling direction; unstable.
Node	Real, same nonzero sign	Negative: attracting; positive: repelling. No spiraling.
Focus (spiral)	$\alpha \pm i\beta$, $\alpha \neq 0$, $\beta \neq 0$	Inward if $\alpha < 0$; outward if $\alpha > 0$.
Center (linear system)	$\pm i\omega$, $\omega > 0$	Closed orbits; stable without attraction.

Negative real parts give local asymptotic stability; a positive real part gives instability. Imaginary or zero eigenvalues require nonlinear analysis. Repeated real roots need eigenvectors to determine the node subtype.

1.5 Equilibria and local linearization

Four local portraits: follow the arrows



Canonical examples, with primes denoting dimensionless time:

Saddle: $q' = q$, $p' = -p$. Node: $q' = -q$, $p' = -2p$.

Focus: $q' = -0.3q + p$, $p' = -q - 0.3p$. Center: $q' = p$, $p' = -q$.

These are local state coordinates, not necessarily displacement and velocity. Reverse the arrows for an unstable node or focus. A center has a family of surrounding periodic orbits.

1.5 Equilibria and local linearization

Worked classification: our oscillator examples

For the perturbation state $\mathbf{y} = (\xi, \dot{\xi})$,

$$A = \begin{pmatrix} 0 & 1 \\ -k_t/m & -c/m \end{pmatrix}, \quad D = c^2 - 4mk_t.$$

For $k_t, c > 0$: $D > 0$ gives a stable node; $D < 0$ gives a stable focus.

$D = 0$: a repeated-root stable node (critical damping).

Classify the cubic equilibria and both pendulum orientations.

Equilibrium	Eigenvalues and type
Cubic, $x_e = 0$	$1.80998, -2.20998 \text{ s}^{-1}$: saddle.
Cubic, $x_e = \pm 0.20 \text{ m}$	$(-0.20 \pm 2.82135 \text{ i}) \text{ s}^{-1}$: stable foci.
Undamped pendulum, down	$\pm i\sqrt{g/L}$: center, justified by energy.
Undamped pendulum, up	$\pm \sqrt{g/L}$: saddle.

1.5 Equilibria and local linearization

Saddle-node: a bifurcation, not an equilibrium type

A saddle and a node meet and disappear as a parameter changes.

$$q' = \mu - q^2, \quad p' = -p, \quad \mu \in \mathbb{R}.$$

Find the equilibria and evaluate the Jacobian at each.

$$p_e = 0, \quad q_e^2 = \mu, \quad A = \begin{pmatrix} -2q_e & 0 \\ 0 & -1 \end{pmatrix}.$$

$\mu > 0$: $q_e = +\sqrt{\mu}$ is a stable node; $q_e = -\sqrt{\mu}$ is a saddle.

$\mu = 0$: the equilibria meet; eigenvalues $0, -1$. Retain $q' = -q^2$.

$\mu < 0$: no equilibrium.

At $\mu = 0$, $q > 0$ approaches zero while $q < 0$ moves away: no stability in a full neighborhood. A zero eigenvalue alone does not prove a saddle-node; the nonlinear terms and parameter dependence matter.

1.6 Phase-plane formulation and trajectory geometry

One second-order equation, two state equations

$$m\ddot{x} + c\dot{x} + r(x) = F_0, \quad v = \dot{x}.$$

Step 1 Choose displacement and physical velocity as the state

Write an equation for each component of the state (x, v) .

$$\dot{x} = v, \quad \dot{v} = \frac{F_0 - cv - r(x)}{m}.$$

Step 2 Find a stationary state

$$\dot{x} = \dot{v} = 0 \implies (x, v) = (x_e, 0), \quad r(x_e) = F_0.$$

A trajectory traces $(x(t), v(t))$ as time increases. An arrow gives the local direction of motion; an equilibrium is a point where both components vanish.

1.6 Phase-plane formulation and trajectory geometry

Nullclines: locate horizontal and vertical motion

$$\dot{x} = v, \quad m\dot{v} = F_0 - cv - r(x).$$

Condition	Direction of motion in the (x, v) plane
$v > 0$ / $v < 0$	Right / left.
$F_0 - cv - r(x) > 0$ / < 0	Up / down.
$v = 0, \dot{v} \neq 0$	Vertical tangent.
$F_0 - cv - r(x) = 0, v \neq 0$	Horizontal tangent.

What happens where both nullclines intersect?

Both derivatives vanish: an equilibrium. A nullcline sets one component to zero; it is generally not a trajectory.

1.6 Phase-plane formulation and trajectory geometry

Worked direction check: reuse the cubic oscillator

For $m = 1$ kg, $c = 0.40$ N s/m, $r(x) = -4x + 100x^3$ N and $F_0 = 0$:

$$\dot{x} = v, \quad \dot{v} = -0.40v + 4x - 100x^3 \quad (\text{SI units}).$$

Evaluate the state velocity at $(x, v) = (0.10 \text{ m}, 0.20 \text{ m/s})$.

$$r(0.10) = -0.30 \text{ N}, \quad \dot{x} = 0.20 \text{ m/s}, \quad \dot{v} = 0.22 \text{ m/s}^2.$$

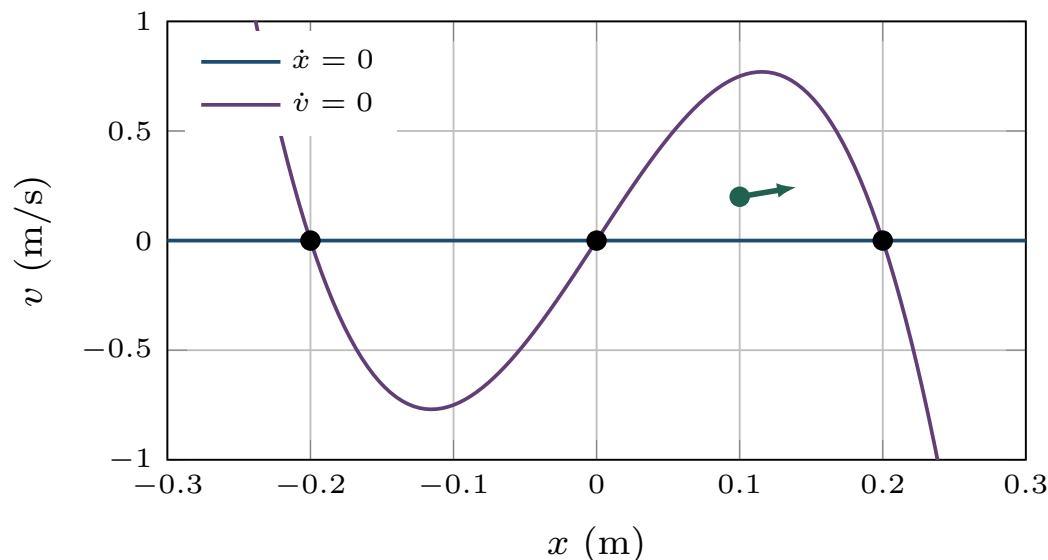
Right and up; compute the trajectory slope where $v \neq 0$

$$\boxed{\frac{dv}{dx} = \frac{\dot{v}}{\dot{x}} = \frac{F_0 - cv - r(x)}{mv}}, \quad \frac{dv}{dx} = 1.10 \text{ s}^{-1}.$$

At $(0.10 \text{ m}, 0)$, $\dot{v} = 0.30 \text{ m/s}^2$: motion is vertically upward. Do not divide by $v = 0$. The derivative dv/dx is with respect to displacement, not time.

1.6 Phase-plane formulation and trajectory geometry

Cubic example: read the nullcline map



Horizontal nullcline

$v = 0$: up or down, except at equilibria.

Curved nullcline

$v = 10x - 250x^3$ (SI): left or right.

Assign directions

Above the curve: $\dot{v} < 0$.

Below the curve: $\dot{v} > 0$.

Above $v = 0$: $\dot{x} > 0$.

Below $v = 0$: $\dot{x} < 0$.

Black points: equilibria. Green: the worked direction check. These curves are nullclines, not solution paths. Axis scales change the apparent arrow angle.

1.6 Phase-plane formulation and trajectory geometry

Why autonomous trajectories cannot cross

At a regular state, one state gives one future

For a locally Lipschitz autonomous field $\dot{\mathbf{z}} = \mathbf{f}(\mathbf{z})$, uniqueness prevents distinct trajectories from crossing.

Can the same (x, v) have two different accelerations in a periodically forced system?

$$m\dot{v} = F \cos \phi - cv - r(x), \quad \dot{x} = v.$$

Yes: the forcing phase ϕ may differ. The pair (x, v) alone does not specify the full state.

Include phase to obtain an autonomous system

$$\dot{x} = v, \quad \dot{v} = \frac{F \cos \phi - cv - r(x)}{m}, \quad \dot{\phi} = \Omega.$$

Here $\phi = \Omega t \pmod{2\pi}$. Uniqueness applies in (x, v, ϕ) ; its projection onto the (x, v) plane can cross. A closed nonstationary orbit of an autonomous system repeats periodically.

1.6 Phase-plane formulation and trajectory geometry

Solved example 12: the undamped center

$$u'' + 2\zeta u' + u = 0, \quad w = u', \quad u' = w, \quad w' = -u - 2\zeta w.$$

Primes denote derivatives with respect to dimensionless time τ .

Set $\zeta = 0$ and release from $(u, w) = (1, 0)$

Find $u(\tau)$ and $w(\tau)$, then eliminate τ .

$$u = \cos \tau, \quad w = -\sin \tau, \quad u^2 + w^2 = 1.$$

$$\mathcal{E} = \frac{1}{2}(u^2 + w^2), \quad \mathcal{E}' = uu' + ww' = 0.$$

At $(1, 0)$ the arrow points down: motion is clockwise. Every nonzero radius gives another closed orbit. The origin is a center: stable, without attraction.

1.6 Phase-plane formulation and trajectory geometry

Damped release: solve for the inward spiral

For $\zeta = 0.20$, $(u(0), w(0)) = (1, 0)$ and $\omega_d = \sqrt{1 - \zeta^2} = \sqrt{0.96}$:

$$u = e^{-\zeta\tau} \left[\cos(\omega_d\tau) + \frac{\zeta}{\omega_d} \sin(\omega_d\tau) \right].$$

Differentiate u to find w . Check both initial conditions.

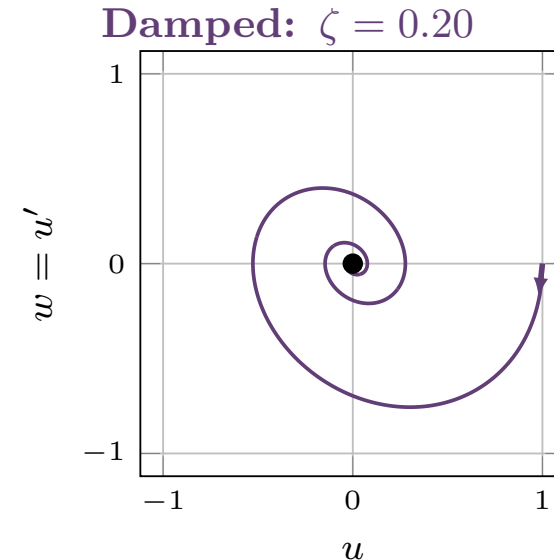
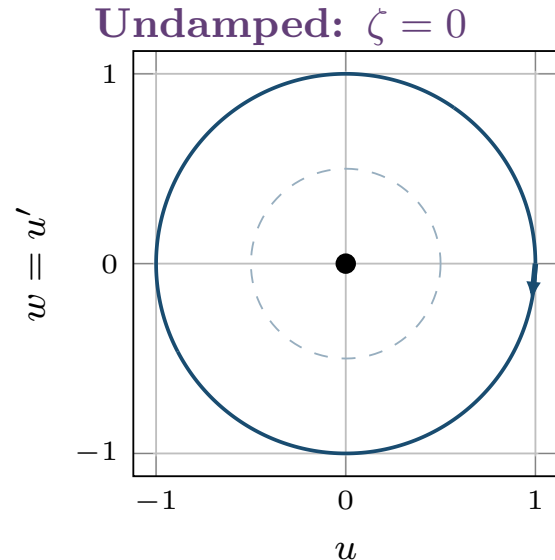
$$w = -\frac{e^{-\zeta\tau}}{\omega_d} \sin(\omega_d\tau).$$

$$\mathcal{E}' = uu' + ww' = -2\zeta w^2 \leq 0, \quad \lambda_{\pm} = -0.20 \pm i\sqrt{0.96}.$$

Both coordinates tend to zero: an inward spiral. At $w = 0$ the energy rate is instantaneously zero, although energy is not conserved along the motion.

1.6 Phase-plane formulation and trajectory geometry

Compare trajectories: center and inward spiral



Solid curves start from $(1,0)$; arrows indicate increasing τ . The dashed circle is another undamped orbit. Equal axis scales make the undamped orbits circular. At $w = 0$, nonstationary paths have vertical tangents.

1.6 Phase-plane formulation and trajectory geometry

A closed orbit is not necessarily a limit cycle

A limit cycle is an isolated periodic orbit

The undamped circles form a continuous family. None is isolated, so they are not limit cycles.

For $\zeta > 0$, rule out nontrivial periodic motion

Assume a period $P > 0$ and integrate the energy rate:

$$\text{Use } \mathcal{E}(P) = \mathcal{E}(0) \text{ and } \mathcal{E}' = -2\zeta w^2.$$

$$0 = -2\zeta \int_0^P w^2 d\tau \implies w \equiv 0.$$

Then $w' = -u - 2\zeta w = 0$ implies $u \equiv 0$.

Only the equilibrium remains. This damped linear oscillator has no nontrivial periodic orbit and hence no limit cycle.

Next, in 1.7: use energy levels to organize nonlinear trajectories and identify separatrices.

1.7 Energy landscapes, turning points and separatrices

From a conserved energy to a phase-plane curve

$$\star \quad m\ddot{x} + \frac{dU}{dx} = 0, \quad v = \dot{x}, \quad m > 0.$$

$$\int m\ddot{x}\dot{x} + \frac{dU}{dx}\dot{x} dt$$
$$\frac{1}{2}m\dot{x}^2 + U(x) = E$$

Multiply by velocity and integrate in time

$$\frac{d}{dt} \left[\frac{1}{2}mv^2 + U(x) \right] = 0,$$

$$E = \frac{1}{2}mv^2 + U(x) = \text{constant.}$$

$$\frac{1}{2}mv^2 + U(x) = E \rightarrow v = \sqrt{\frac{2}{m}(E - U(x))}$$

$$v = \pm \sqrt{\frac{2}{m}[E - U(x)]}, \quad U(x) \leq E.$$

Upper branch: rightward motion. Lower branch: leftward motion. The trajectory stays in the connected accessible region containing its initial state.

U is time independent. Include a constant load through $U = V - F_0x$.

1.7 Energy landscapes, turning points and separatrices

Turning points and the period integral

At a candidate turning point, $U(x) = E$ and $v = 0$.

If $dU/dx \neq 0$, acceleration is nonzero and motion reverses. If also $dU/dx = 0$, the state is an equilibrium; it is not an ordinary turning point.

For oscillation between two simple turning points

$$dt = \sqrt{\frac{m}{2}} \frac{dx}{\sqrt{E - U(x)}} \quad (v > 0),$$

$$T(E) = \sqrt{2m} \int_{x_-}^{x_+} \frac{dx}{\sqrt{E - U(x)}}.$$

For $U = \frac{1}{2}kx^2$, $E = \frac{1}{2}kA^2$, substitute $x = A \sin \psi$.

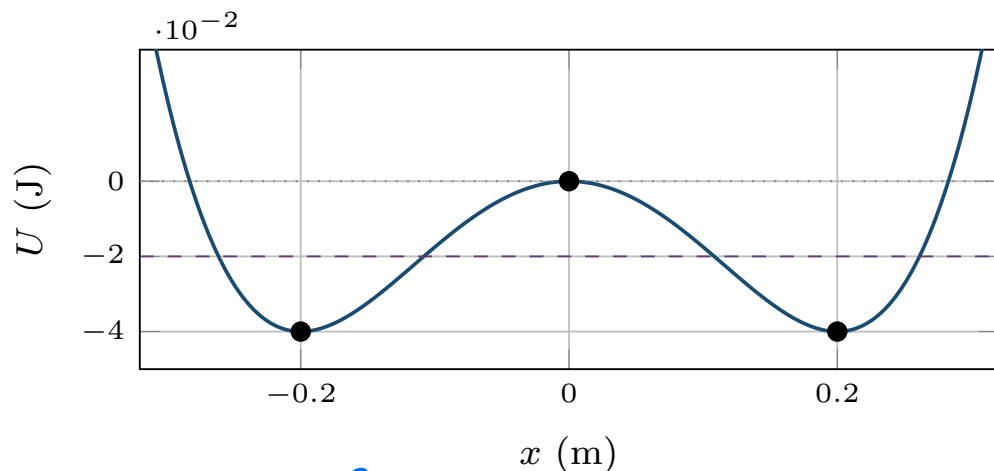
$$T = 2\sqrt{m/k} \int_{-\pi/2}^{\pi/2} d\psi = \boxed{2\pi\sqrt{m/k}}.$$

Double the left-to-right travel time; require $E > U(x)$ inside. In general, T depends on E .

1.7 Energy landscapes, turning points and separatrices

Solved example 13: the double-well energy barrier

Set $c = 0$ in Example 10: $m\ddot{x} - ax + bx^3 = 0$, with $m = 1$ kg, $a = 4$ N/m, $b = 100$ N/m³.
 $U(x) = -\frac{1}{2}ax^2 + \frac{1}{4}bx^4 = -2x^2 + 25x^4$ (SI units).



Minima and barrier

$$U(\pm 0.20) = -0.0400 \text{ J.}$$

$$U(0) = 0.$$

$$\Delta U = 0.0400 \text{ J.}$$

Dashed line: $E = -0.0200$ J.

Energy zero is arbitrary; the barrier height is an energy difference.

$$r(x) = -ax + bx^3 \rightarrow U(x) = \int -ax + bx^3 dx = -\frac{1}{2}ax^2 + \frac{1}{4}bx^4 \quad \text{I can plot}$$

$$\frac{1}{2}mv^2 + U(x) = E \rightarrow v = \pm \sqrt{\frac{2}{m} \left[E + \frac{1}{2}ax^2 - \frac{1}{4}bx^4 \right]} \rightarrow \text{U-x phase for different E}$$

1.7 Energy landscapes, turning points and separatrices

Double-well example: calculate trapped motion

Take $E = -0.0200$ J and an initial state in the right well.

Find the two turning points

$$25x^4 - 2x^2 = E, \quad y = x^2, \quad y = \frac{2 \pm \sqrt{4 + 100E}}{50} \quad (\text{SI units}).$$

Choose the positive roots. Then find the speed at $x = 0.20$ m.

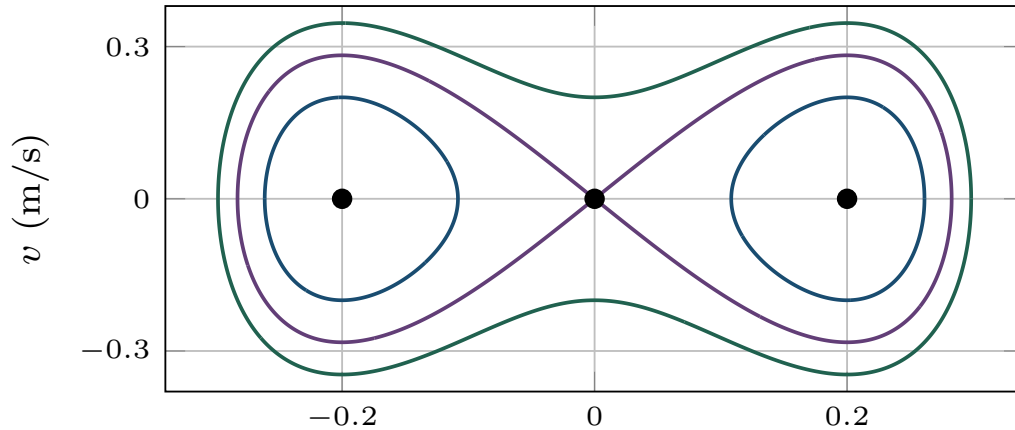
$$x_- = 0.108239 \text{ m}, \quad x_+ = 0.261313 \text{ m}.$$

$$|v| = \sqrt{\frac{2}{m}[E - U(0.20)]} = 0.2000 \text{ m/s}.$$

The left-well interval is $[-x_+, -x_-]$. At this energy, $x = 0$ is inaccessible because $U(0) > E$; motion stays in its initial well.

1.7 Energy landscapes, turning points and separatrices

The separatrix separates trapped and cross-well motion



$$v = \pm \sqrt{\frac{2}{m} \left[E + \frac{1}{2} a x^2 - \frac{1}{4} b x^4 \right]}$$

Upper branches move right; lower branches move left. From a well bottom, the threshold speed is $\sqrt{2\Delta U/m} = 0.282843$ m/s. Exactly at threshold, the saddle is approached in infinite time; crossing needs strictly more energy.

$$E=0 \rightarrow v = \pm \sqrt{\frac{2}{m} \left[\frac{1}{2} a x^2 - \frac{1}{4} b x^4 \right]} \rightarrow \text{plot}$$

$-0.04 < E < 0$: one well.

$E = 0$: separatrix lobes.

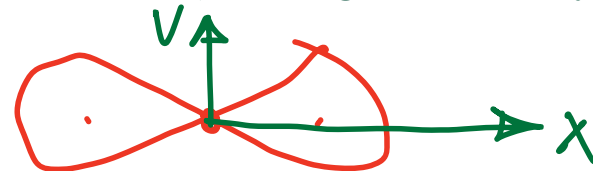
$E > 0$: both wells.

At the barrier

$$v = \pm x \sqrt{\frac{a}{m} - \frac{b}{2m} x^2},$$

$$|x| \leq \sqrt{2a/b} = 0.282843 \text{ m.}$$

Each lobe approaches the same saddle in both time directions: a homoclinic orbit. It is not periodic.



1.7 Energy landscapes, turning points and separatrices

Solved example 14: normalize the pendulum energy

$$E_{\text{phys}} = \frac{1}{2}mL^2\dot{\theta}^2 + mgL(1 - \cos\theta).$$

Use $\tau = \sqrt{g/L}t$, $w = \theta' = d\theta/d\tau$ and $\dot{\theta} = \sqrt{g/L}w$.

$$\frac{1}{2}\dot{\theta}^2 + \frac{g}{L}\cos\theta = E \rightarrow \text{datum}$$

$$V_g = mgL(1 - \cos\theta)$$

$$\mathcal{E} = \frac{E_{\text{phys}}}{mgL} = \frac{1}{2}w^2 + 1 - \cos\theta.$$

$$\mathcal{E}' = w(\theta'' + \sin\theta) = 0.$$

The downward position has potential energy 0; the upright position has potential energy 2. Thus $\mathcal{E} = 2$ is the barrier level.

Dots denote physical-time derivatives; primes denote dimensionless-time derivatives. The physical barrier is $2mgL$.

$$mL^2\ddot{\theta} + mLg\sin\theta = 0$$

$$\int mL^2\ddot{\theta} + mLg\sin\theta \, dt$$

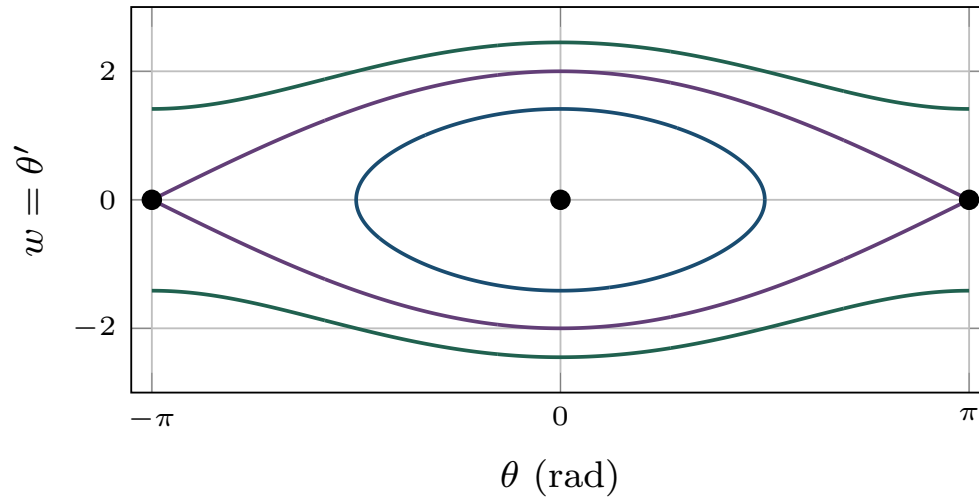
$$\frac{1}{2}mL^2\dot{\theta}^2 + mgL\cos\theta = E$$

$$\rightarrow \mathcal{E} = \frac{1}{2}\dot{\theta}^2 + 1 - \cos\theta$$

$$\dot{\theta} = \pm \sqrt{2(\cos\theta - 1) + 2\mathcal{E}}$$

1.7 Energy landscapes, turning points and separatrices

Pendulum energy levels: libration or rotation



At $\theta = 0$, $w = 1$: $\mathcal{E} = 0.5$, so $\theta_{\max} = \pi/3 = 60^\circ$.

$\mathcal{E} = 0$: downward rest.

$0 < \mathcal{E} < 2$: libration.

$$\theta_{\max} = \arccos(1 - \mathcal{E}).$$

$\mathcal{E} = 2$: separatrices and upright equilibria.

$\mathcal{E} > 2$: rotation; w never vanishes.

Curves shown: $\mathcal{E} = 1, 2, 3$. Upper branches move toward increasing θ .

1.7 Energy landscapes, turning points and separatrices

Pendulum threshold: why it takes infinite time

At $\mathcal{E} = 2$ and $-\pi \leq \theta \leq \pi$:

$$w^2 = 2(1 + \cos \theta) \implies \boxed{w = \pm 2 \cos(\theta/2)}.$$

$$|\dot{\theta}(0)|_{\text{threshold}} = 2\sqrt{g/L} = 6.26418 \text{ rad/s} \quad (L = 1 \text{ m}).$$

Approach the upright position on the upper branch

Let $\delta = \pi - \theta > 0$. Then $w = 2 \sin(\delta/2) \sim \delta$.

Use $d\tau = d\theta/w$ to determine whether the arrival time is finite.

$$d\tau \sim -\frac{d\delta}{\delta} \implies \tau \rightarrow \infty \text{ as } \delta \rightarrow 0.$$

Exactly at threshold: asymptotic approach, not finite-period motion. On the unwrapped plane a branch connects adjacent saddles (heteroclinic); identifying θ modulo 2π gives the same saddle on a cylinder (homoclinic). Here $g = 9.81 \text{ m/s}^2$.

WEEK 1 REFERENCES

Sources and their role in this lecture

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Secs. 1.2 and 2.1.1–2.1.2: conservative oscillators, pendulum and nonlinear restoring forces.

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Harmonic Balance for Nonlinear Vibration Problems. Springer, 2019.

Sec. 1.2: Duffing oscillator, superposition limits and Fourier representation; Ch. 2 supports later harmonic-balance lectures.

[3] O. Thomas

“Understanding, computing and identifying the nonlinear dynamics of elastic and piezoelectric structures thanks to nonlinear modes.” In *Model Order Reduction for Design, Analysis and Control of Nonlinear Vibratory Systems*, 2024.

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[4] C. Touzé and A. Vizzaccaro

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Sec. 2: mechanical models, quadratic/cubic terms and state-space formulation. DOI: 10.1007/978-3-031-67499-0_2.

Equations and vector figures are newly typeset. Worked numerical examples and supporting calculations are original teaching derivations using these model families.